

Automatic GFL/GFM Mode Switching

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Introduction

Grid-following (GFL)

- PLL-synchronized current control
- Requires an external voltage-angle reference
- Effective while a strong grid reference exists

Grid-forming (GFM)

- Establishes voltage angle and frequency
- Supports weak-grid and islanded operation
- Can replace the lost reference after SG disconnection

Research problem

Automatically assign only the required GFM support after loss of the synchronous reference, then return converters to GFL operation after SG recovery.

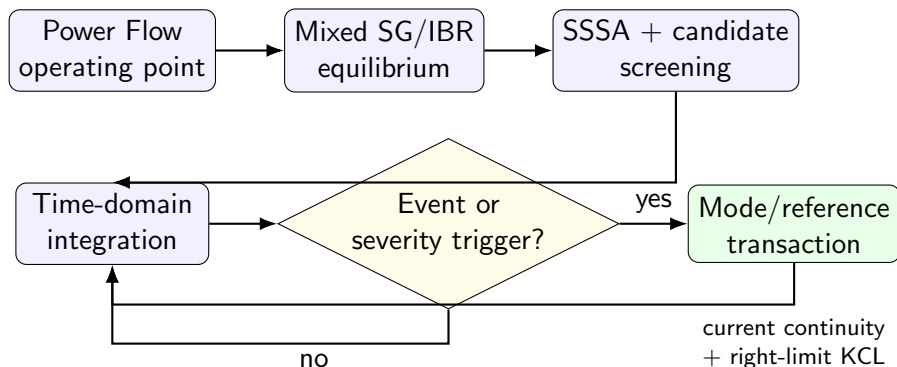
Project Objectives

- 1 Develop one in-house analysis chain for **Power Flow**, **SSSA**, and **Time-Domain Simulation**.
- 2 Model four converter resources with a common plant and selectable **GFL/GFM** controller branches.
- 3 Determine admissible GFM configurations from the network condition instead of fixing the number of forming units in advance.
- 4 Verify automatic mode transitions, island support, and SG reconnection under one disturbance chronology.

Scope

IEEE 14-bus test system, one synchronous generator, four dual-mode IBRs, automatic reference support, and event-driven nonlinear simulation.

Overall Analysis and Switching Workflow



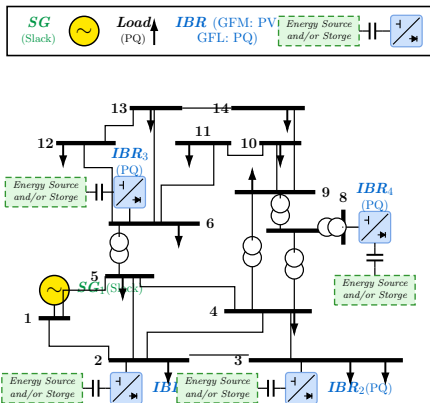
Key principle: every mode change is checked against a certified operating point and committed only if the post-transition network equations remain solvable.

Case Study: IEEE 14-Bus with 1 SG + 4 Dual-Mode IBRs

Resource mapping

Resource	Connection
SG1	1
IBR1	2
IBR2	3
IBR3	6
IBR4	8

- SG1: normal synchronous reference
- IBR1–IBR4: dual-mode GFL/GFM converters



Study question: Which IBRs should form the grid after SG1 is lost, and when can that support be released after SG1 returns?

Power Flow Formulation

For network node i ,

$$P_i = V_i \sum_{k=1}^n V_k (G_{ik} \cos \theta_{ik} + B_{ik} \sin \theta_{ik}),$$

$$Q_i = V_i \sum_{k=1}^n V_k (G_{ik} \sin \theta_{ik} - B_{ik} \cos \theta_{ik}).$$

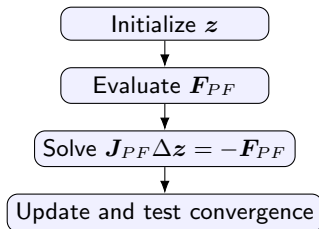
Newton–Raphson update

$$\mathbf{J}_{PF}(\mathbf{z}^{(r)}) \Delta \mathbf{z}^{(r)} = -\mathbf{F}_{PF}(\mathbf{z}^{(r)})$$

$$\mathbf{z}^{(r+1)} = \mathbf{z}^{(r)} + \Delta \mathbf{z}^{(r)}$$

Network constraints

Type	Specified	Solved
REF	$ V , \theta$	P, Q
PV	$P, V $	Q, θ
PQ	P, Q	$ V , \theta$



Small-Signal Stability Analysis (SSSA)

Nonlinear DAE:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{y}), \quad \mathbf{0} = \mathbf{g}(\mathbf{x}, \mathbf{y})$$

Linearization around $(\mathbf{x}_0, \mathbf{y}_0)$:

$$\begin{bmatrix} \Delta \dot{\mathbf{x}} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_x & \mathbf{f}_y \\ \mathbf{g}_x & \mathbf{g}_y \end{bmatrix} \begin{bmatrix} \Delta \mathbf{x} \\ \Delta \mathbf{y} \end{bmatrix}$$

Eliminating algebraic variables:

$$\boxed{\mathbf{A} = \mathbf{f}_x - \mathbf{f}_y(\mathbf{g}_y \backslash \mathbf{g}_x)}$$

$$\boxed{\mathbf{A}\mathbf{v} = \lambda\mathbf{v}}$$

$$f = \frac{|\operatorname{Im} \lambda|}{2\pi}, \quad \zeta = -\frac{\operatorname{Re} \lambda}{|\lambda|}$$

Stability criterion

$\operatorname{Re}(\lambda) < 0$ for all physical modes; the selector also checks the declared damping-ratio floor.

Time-Domain Simulation (TS)

$$\dot{x} = f(x, y), \quad 0 = g(x, y)$$

Implicit trapezoidal step

$$R_x = x_{n+1} - x_n - \frac{h}{2}(f_n + f_{n+1}) = 0$$

$$R_y = g(x_{n+1}, y_{n+1}) = 0,$$

$$\boxed{J_R \Delta z} = \boxed{-R}, \quad z = [x; y].$$

Event handling

- Exact landing on event times
- Preserve the left-limit state
- Atomic mode/topology transaction
- Right-limit KCL must converge
- Roll back a rejected transaction

Dual-Mode IBR Dynamic Model

Fixed 17-coordinate switching superset

$$\mathbf{x}_{IBR} = [\mathbf{x}_p^T \quad \mathbf{x}_{GFL}^T \quad \mathbf{x}_{GFM}^T \quad I_{dc}]^T$$

$$\mathbf{x}_p = [i_d \ i_q \ V_{dc}]^T,$$

$$\mathbf{x}_{GFL} = [\delta_{PLL} \ \xi_{PLL} \ \xi_P \ \xi_Q \ \xi_{Id} \ \xi_{Iq}]^T,$$

$$\mathbf{x}_{GFM} = [\delta_{VSG} \ \omega_{VSG} \ E \ \xi_{Vd} \ \xi_{Vq} \ \xi_{Id}^{GFM} \ \xi_{Iq}^{GFM}]^T$$

$$\mathcal{A}_{GFL} = \{1:9, 17\}, \quad \mathcal{A}_{GFM} = \{1:3, 10:17\}.$$

Inactive controller coordinates are held during each mode.

GFL: PLL-synchronized

$$\dot{\delta}_{PLL} = k_{p,PLL} v_q + k_{i,PLL} \xi_{PLL},$$

$$e_P = \kappa P^* - P,$$

$$i_d^* = k_p P e_P + k_i P \xi_P.$$

GFM: voltage-forming VSG

$$M\dot{\omega} = \kappa P^* - P - D_v(\omega - 1),$$

$$\dot{\delta}_{VSG} = \omega_b(\omega - 1),$$

$$\tau_E \dot{E} = k_Q(\kappa Q^* - Q) - k_E(E - E^*).$$

Automatic GFL/GFM Switching Law

Severity index

$$J_V = \frac{||V_i| - V_i^*|}{0.10 \text{ p.u.}},$$

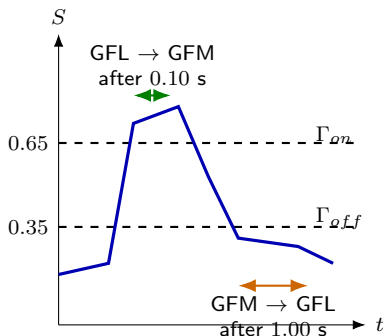
$$J_f = \frac{|f_{COI} - 60|}{0.50 \text{ Hz}},$$

$$S = \text{sat}_{[0,1]}(0.5J_V + 0.5J_f).$$

Hysteretic mode command

$$q^+ = \begin{cases} \text{GFM}, & S \geq 0.65 \quad (0.10 \text{ s}), \\ \text{GFL}, & S < 0.35 \quad (1.00 \text{ s}), \\ q, & \text{otherwise.} \end{cases}$$

$$\|I^+ - I^-\|_\infty \leq 10^{-10} \text{ p.u.}$$



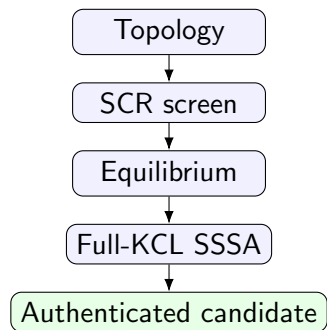
Candidate equilibrium/SSSA checks and the right-limit KCL solve are completed before a transition is committed.

Result 1: GFM Candidate Screening

GFM count	Candidate	Outcome
1	any single IBR	admissible
2	IBR1 + IBR3	admissible
2	IBR2 + IBR3	best margin
2	other pairs	rejected
3	any triple	rejected
4	all four IBRs	admissible

Observed property

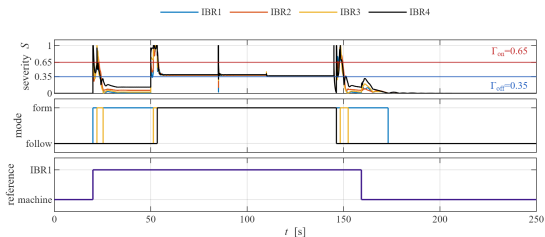
Admissibility is **non-monotonic** in the number of grid-forming converters.



The GFM set is computed from the current network condition; converter count alone does not determine admissibility.

Result 2: Supervisor and Mode Switching

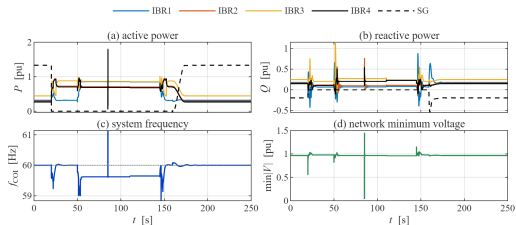
- SG trip: 20 s
- Load step: 50 s
- Bus-9 fault: 85–85.15 s
- Line 6–13 trip: 110 s
- Restore / SG reclose request: 145 s



Policy: add GFM support during severe conditions and release it after recovery.

Result 3: Electrical Response

- IBRs support the island after the SG trip.
- Frequency and voltage respond to each scheduled disturbance.
- The SG is reconnected only after the synchronism conditions are satisfied.
- The accepted run reaches the full 250 s horizon.

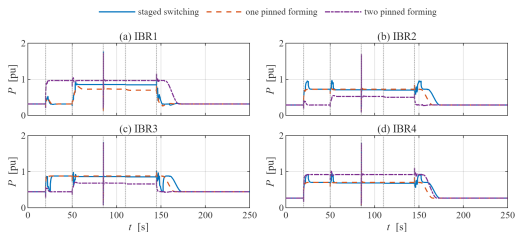


Result 4: Converter Response under Different Policies

- Staged switching: solid lines
- One pinned GFM: dashed lines
- Two pinned GFMs: dash-dot lines

Main observation

The power-sharing trajectory depends strongly on which converters are forming the grid and on when each mode transition occurs.



Conclusion and Key Findings

- PF, SSSA, and TS are linked through one nonlinear SG/IBR network model.
- After SG1 is disconnected, at least one voltage-forming IBR is required for the islanded trajectory to continue.
- GFM admissibility is configuration-dependent and non-monotonic in converter count.
- Staged switching reaches the full 250 s chronology and returns all IBRs to GFL operation after SG recovery.

Scope of claim: the staged policy is not claimed to minimize every transient peak; its benefit here is adaptive reference support across the complete disturbance chronology.

Comparative outcome

- **Locked GFL:** stops at 20 s; no forming source.
- **1–2 pinned GFMs:** reach 250 s; reclose succeeds.
- **4 pinned GFMs:** nonlinear failure at 25.485 s.
- **Staged switching:** 250 s with full hand-back.